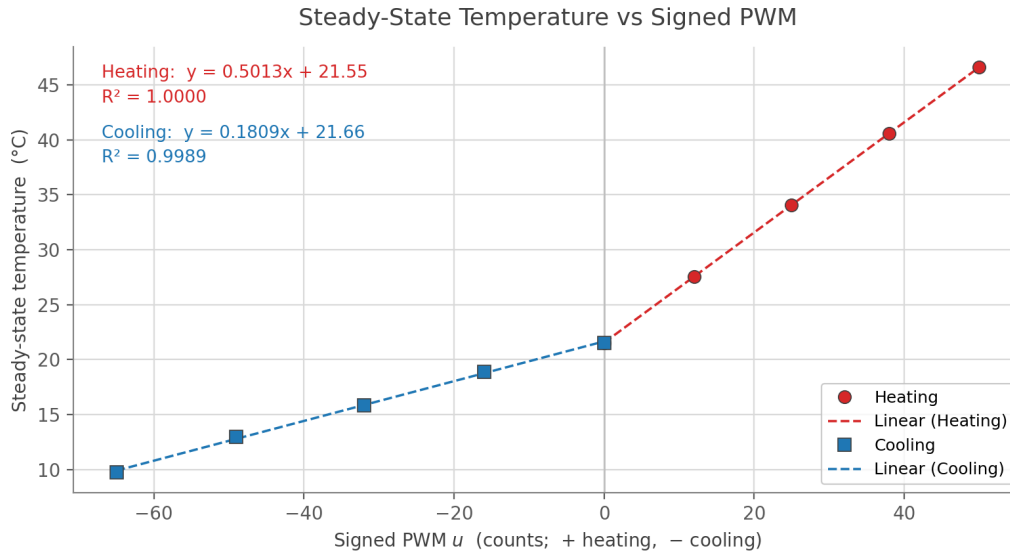


A2: TEC Heating and Cooling Analysis

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1. Steady-state temperature versus signed PWM



Steady state: after each PWM step the trace was watched for about three thermal time constants ($\tau = 70$ s driven), then for one further minute. A point was accepted once its net drift over that minute was no larger than the measured short-term noise, 0.16 °C peak-to-peak. Points that had not fully arrived are reported as the asymptote of an exponential fit to the approach; steady_state.csv records which. $T_0 = 21.54$ °C. Both branches linear: $R^2 = 1.0000$ heating, 0.9989 cooling.

Ten steady-state points, five per branch, about the zero-PWM temperature $T_0 = 21.54$ °C. Steady state: after each PWM step the trace was watched for about three thermal time constants ($\tau \approx 70$ s driven) and then one further minute, and a point was accepted once its net drift over that minute was no larger than the measured short-term noise of 0.16 °C peak-to-peak. Points that had not fully arrived within the session are reported as the asymptote of an exponential fit to the approach; data/module_04/steady_state.csv records which. Trendlines are fitted only over the range of each branch.

2. Measured susceptibilities and their ratio

Branch	Susceptibility dT/du	Fit range (counts)	R^2	Intercept
Heating	$m_h = 0.5013$ °C per PWM count	$u = 0$ to +50	1.0000	21.55 °C
Cooling	$m_c = 0.1809$ °C per PWM count	$u = -65$ to 0	0.9989	21.66 °C

Both branches are linear; neither needs to be reported as a local estimate. $r = m_h/m_c = 2.771$. The two intercepts differ by 0.11 °C and the cooling one sits 0.12 °C above the measured T_0 , most likely the room drifting over a two-hour session.

3. PWM averaging, steady-state balance, and the slope ratio

(a) How PWM averages current. Over one period the current is I for a time $D\tau$ and zero for the remaining $(1-D)\tau$, with $D = |u|/255$. Both integrands vanish off the on-interval, so

$$\langle I \rangle = (1/\tau) \int I dt = (1/\tau) \cdot I \cdot D\tau = DI \quad \langle I^2 \rangle = (1/\tau) \int I^2 dt = (1/\tau) \cdot I^2 \cdot D\tau = DI^2$$

$\langle I^2 \rangle$ is not $\langle I \rangle^2 = D^2 I^2$: squaring and averaging do not commute, and for a two-level waveform they differ by exactly $1/D$. This is what makes the graph straight. Peltier transport follows $\langle I \rangle$ and Joule heating follows $\langle I^2 \rangle$, and both are *linear in D* , so the susceptibility is constant. Had the Joule term gone as D^2 the branches would curve measurably; the heating branch is linear to one part in a thousand, so the measurement supports the result. (With a DAC supplying steady current, $\langle I^2 \rangle = \langle I \rangle^2$ would hold; the distinction is a property of PWM.)

(b) Steady state. From $C dT/dt = Q_{TEC} - G(T - T_0)$ with $dT/dt = 0$, the individual flows are not zero but their sum is, so $G(T - T_0) = Q_{TEC}$. With signed duty $d = u/255$, Peltier reversing with the current and Joule heating not: $Q_{TEC} = dQ_p + |d|Q_j$. On the heating branch $d > 0$ so both terms deliver heat into the object and conduction carries it out; on the cooling branch $d < 0$, Peltier removes heat while Joule still adds it, and conduction supplies heat inward. Hence

$$\begin{aligned} T_h - T_0 &= d(Q_p + Q_j)/G & T_c - T_0 &= d(Q_p - Q_j)/G \\ dT_h/dd &= (Q_p + Q_j)/G & dT_c/dd &= (Q_p - Q_j)/G \end{aligned}$$

(c) The ratio. Since $d = u/255$, each measured slope carries a factor $1/255$, and that factor and G are common to both, so both cancel from r :

$$r = (Q_p + Q_j)/(Q_p - Q_j) \Rightarrow Q_j/Q_p = (r - 1)/(r + 1) \quad \text{check: } r = 2 \text{ gives exactly } 1/3$$

With $r = 2.771$: $Q_j/Q_p = 0.470$. Joule heat delivered to the object face is about 47% of the Peltier pumping. That r is independent of G , which was never measured, is what makes it the useful quantity here.

4. Laird CP14-127-045 data sheet, hot side 27.0 °C

Quantity	Value	Meaning and the operating condition attached
R_M	1.50 Ω	Ohmic resistance of the 127 couples in series, quoted at $T_h = 27\text{ }^\circ\text{C}$; rises to 1.68 Ω at 50 $^\circ\text{C}$
$I_{\max}$	8.6 A	The current giving the <i>largest temperature difference</i> , not a safety ceiling. Condition: I at $\Delta T_{\max}$
$Q_{c,\max}$	71.3 W	Most heat the cold face can absorb. Condition: $\Delta T = 0$, no difference to pump against
$\Delta T_{\max}$	70.5 $^\circ\text{C}$	Largest sustainable temperature difference. Condition: $Q_c = 0$, no heat load

The last two sit at opposite ends of one trade-off: maximum pumping requires zero temperature difference, maximum difference requires zero load. At the data-sheet condition $\Delta T = 0$, so conduction vanishes and the symmetric model puts half the Joule heat on each face:

$$Q_{J,\max} = \frac{1}{2} I_{\max}^2 R_M = \frac{1}{2} (8.6\text{ A})^2 (1.50\text{ }\Omega) = 55.47\text{ W}$$

$$Q_{P,\max} = Q_{c,\max} + Q_{J,\max} = 71.3\text{ W} + 55.47\text{ W} = 126.77\text{ W}$$

$$r_{\text{Laird},\max} = (126.77 + 55.47) / (126.77 - 55.47) = 2.556$$

Cross-check that the right rows were read: $V_{\max}/I_{\max} = 13.9/8.6 = 1.616\text{ }\Omega$, which is 8% above the quoted R_M . Not an inconsistency — at $\Delta T_{\max}$ the module develops a Seebeck back-EMF, so $V_{\max} = I_{\max} R_M + S \cdot \Delta T_{\max}$, giving $S = 0.0142\text{ V/K}$, a sensible coefficient for 127 bismuth telluride couples.

5. Comparison of measured and data-sheet ratios

	r	Q_J/Q_P	Condition
Measured, this apparatus	2.771	0.470	PWM, finite ΔT , current set by the supply and bridge
Laird, at $I_{\max}$	2.556	0.438	steady DC at 8.6 A, $\Delta T = 0$
ratio	1.084		agreement to 8%

They are not expected to agree, and 8% is closer than this comparison deserves. $D = 1$ does not imply $I = I_{\max}$: full duty only means the bridge is continuously on, while the actual current is set by the 12 V supply and its limit, the H-bridge forward drop, wiring resistance and R_M , which together put the bench current well below 8.6 A. Beyond that: our current is chopped rather than steady DC; the data-sheet relation holds at $\Delta T = 0$ whereas the plate ran 25 $^\circ\text{C}$ above and 12 $^\circ\text{C}$ below ambient, where the conduction term is not negligible and is absorbed into G ; the coefficients move with temperature (the same sheet shows R_M rising 1.50 $\rightarrow$ 1.68 Ω between 27 and 50 $^\circ\text{C}$); and one slope is fitted to each branch. *Direction of the difference*: ours is the larger, yet operating below $I_{\max}$ should reduce Q_J/Q_P , since that ratio scales as I . The opposite sign points at dissipation our model assigns to the object face that the data sheet does not carry, such as lead and contact resistance in series with the module.

6. Passive conduction

When the object is hotter than the room, heat flows **out** of it; when colder, heat flows **in**. Either way the flow drives the object back toward room temperature, so conduction opposes the TEC on both branches, which is why it enters as $-G(T-T_0)$ with a single positive G . It cannot explain the unequal slopes: the *same* G appears in the denominator of both dT_h/dd and dT_c/dd , so it sets how steep the two branches are — a larger G flattens both — but cancels exactly from their ratio. A symmetric term cannot produce an asymmetric result. The asymmetry survives only in the numerators, where Q_J is added on the heating branch and subtracted on the cooling one, because Peltier transport reverses with the current and Joule heating does not.

7. Conclusion

Steady-state temperature is linear in signed PWM on both branches ($R^2 = 1.0000$ heating, 0.9989 cooling), which is what the PWM averages predict: Peltier transport follows $\langle I \rangle = DI$ and Joule heating follows $\langle I^2 \rangle = DI^2$, so both scale with duty and the susceptibility stays constant. Had $\langle I^2 \rangle$ gone as D^2 , the branches would curve. The two slopes differ by a factor $r = 2.771$ because reversing the current reverses Peltier pumping but not Joule heating: the two add when heating and oppose when cooling, giving $Q_J/Q_P = (r-1)/(r+1) = 0.470$. Passive conduction enters both branches through the same G and cancels from the ratio, so it sets how steep the lines are but cannot make them unequal. The Laird maximum-current figures predict $r = 2.556$, within 8% despite describing a steady-DC, zero- ΔT condition our apparatus never reaches.

Code and data: `arduino/module_04/tec_open_loop_safety` (safety interlock), `python/tec_control_gui.py`, `python/plot_open_loop_calibration.py`, `python/laird_comparison.py`; raw traces and the steady-state table in `data/module_04/`.